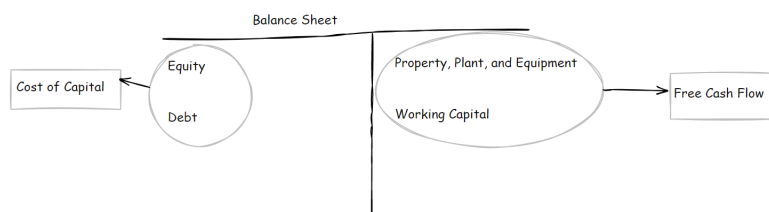


Business Analysis and Valuation

Estimating the Cost of Capital

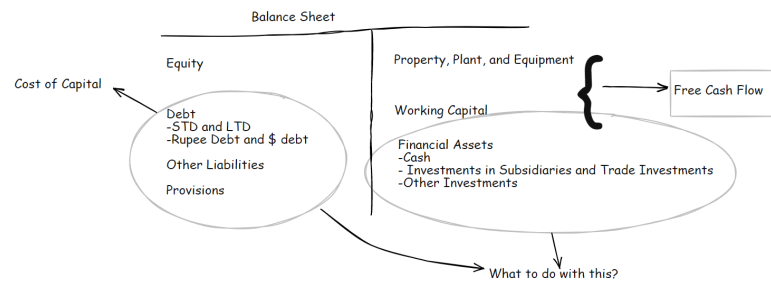
1

A simple Valuation Setting



2

But real life is not simple



3

Look at Titan's Liabilities

EQUITY AND LIABILITIES			
Equity			
(a) Equity share capital	13.1	89	89
(b) Other equity	13.2	20,398	16,722
	TOTAL EQUITY	20,487	16,811
Liabilities			
(1) Non-current liabilities			
(a) Financial liabilities	14.1	-	420
(i) Borrowings	14.2	2,142	1,943
(ii) Lease liabilities	15	213	256
(b) Provisions		2,355	2,619
(2) Current liabilities			
(a) Financial liabilities			
(i) Borrowings	16.1	6,239	7,483
(ii) Gold on loan	16.2	14,314	7,043
(iii) Lease liabilities	16.3	314	282
(iv) Trade payables			
- Total outstanding dues of micro and small enterprises	16.4	531	158
- Total outstanding dues of creditors other than micro and small enterprises	16.4	1,384	1,314
(v) Other financial liabilities	16.5	2,244	1,134
(b) Other current liabilities	17	5,466	4,069
(c) Provisions	18	269	130
(d) Current tax liabilities (net)	8	14	32
		30,775	21,645
TOTAL EQUITY AND LIABILITIES		53,617	41,075

4

Two questions

- When we estimate the cost of capital, which figures will be included in debt?
- When we find the enterprise value, what must be subtracted from it to get the equity value?

5

Four Technical Points about Cost of Capital

- **Pre-tax cost of debt is not the same thing as the coupon rate of the bond.**
- **Cost of equity is not the same as dividend yield.**
- **The weights in the computation of cost of capital must be based on market values of debt and equity and not their book values.**
- **The discount rate must match the riskiness and the type of cash flows being discounted.**

6

What if pre-tax cost of debt is different from the coupon rate?

- It does not matter if the growth rate is 0.
- Let's take an example:

• Earnings before Interest and Tax	Rs. 100 millions
• Book Value of Debt	Rs. 100 millions
• Tax Rate	30%
• Cost of Equity	20%
• Yield to maturity	12%
• Coupon rate	10%

7

Solution

Use Coupon Rate

- EBIT = 100 million
- Interest: 10 (=100*10%)
- PBT: 90
- Tax: 27
- PAT: 63
- Equity: $63/0.2 = 315$
- FCF: 70
- $WACC = \frac{315}{425} \times 0.2 + \frac{100}{415} \times 0.07 = 16.87\%$
- Ent Value = $\frac{70}{16.87\%} = 415.$

Use YTM

- EBIT = 100 million
- Interest: 10 (=83.33*12%)
- PBT: 90
- Tax: 27
- PAT: 63
- Equity: $63/0.2 = 315$
- FCF: 70
- $WACC = \frac{315}{398.33} \times 0.2 + \frac{83.33}{398.33} \times 0.084 = 17.57\%$
- Ent Value = $\frac{70}{17.57\%} = 398.33.$

What is the cost of capital of the company? 16.87% or 17.57%?

8

An example

- A project with an initial investment of Rs.500 millions and an IRR of 17% is available for the above company. Should the company invest in the project?
 - Its cost of capital is 16.87% (using coupon rate) or 17.57% (using cost of debt).
- Assume that we decide to use 16.87% as the cost of capital.
- So, we decide to invest in the project.
- We also raise Rs.500 millions using the current DE to keep the cost of capital constant.

9

The Solution

Raising Rs.500 Million	
Total Capital	500
Current Debt	100
Current Equity	315.00
Extra Debt	120.48
Extra Equity	379.52
IRR	17%

After the Investment	
EBIT-old	100
EBIT-Project	121.43
EBIT-New	221.43
Interest-old	10
Extra Interest	14.46
Interest-New	24.46
PBT	196.97
Tax	59.09
PAT	137.88
Equity	689.40
NPV	-5.12

10

Weights used to compute WACC

- Why do we use market-value based weights?
 - Book value per share of Titan is Rs.230.19. Its shares are trading at around Rs.4,629.
 - Titan's cost of equity is about 12%.
- X invested Rs.100 in 2002 in a company. Today, the book value of his investment is Rs.300. But the market value of that investment is Rs.1,000. Why should X bother about the market value of the investment?

11

Match the Riskiness and Type of Cash Flows

- Match the Riskiness
 - Equity Cash flows discounted at K_e . Free Cash Flows discounted at WACC.
- Match the Type
 - Nominal cash flows discounted at nominal cost of capital.
 - Rupee cash flows discounted at rupee cost of capital.

12

Real and Nominal Rates

- Assume that the real rate of interest is 3%.
- Also, assume the inflation rate to be 6%.
- Then the nominal rate of interest is 9.18%.
- Why?
- Assume that everyone buys only tomato. Tomato costs Rs.100 a KG today.
- Tomato is expected to cost Rs.106 next year.
- Therefore, I will deposit Rs.100 in a bank today only if the bank gives me **** next year.

13

Which cashflow to use - Nominal or Real Cash Flows?

- Assume that ASL Limited sells 1,000 units of JLTs each year.
- The unit selling price is Rs.100 and the unit cost of production (all direct costs) is Rs.60.
- Assume that the real discount rate is 3% and the inflation rate is 4%.
- What is the value of ASL?
- Assumptions made

14

Interest Rate Parity

- The exchange rate between \$ and ₹ is ₹80 per \$.
- Interest rate in the U.S. is 5% p.a.
- Interest rate in India is 7% p.a.
- If you have \$100, where will you invest?

15

Dollar or Rupee Cash Flows...

- An American investor wants to invest in ASL (previous example). Assume that the US real discount rate is 3% and the inflation rate is 1%. Assume that the spot exchange rate is Rs.80/\$.
- What is the value of ASL to the American investor?
- Assumptions made in IRP

16

In summary...

- Use real or nominal discount rate?
- Use \$ or ₹ discount rate?

17

Estimating Cost of Debt

- The Pecking Order
 - Find the YTM of the Bond (Pecking Order 1)
 - Ask someone who knows (Pecking Order 2)
 - Can use Bloomberg here
 - Use the Credit Rating of the Bonds issued by the Company (Pecking Order 3)
 - Rate the Bonds Yourself (Pecking Order 4)

18

Debt of Indian Companies

- 83 of the 500 companies (BSE 500) have borrowed directly from the public as on FY 2023.
- 146 companies have borrowed from the banks directly.

19

Credit Rating and Yield Spread...

- Data on corporate yield spread not available from any source.
- FIMMDA shares some data with the general public.

Annualised spreads	0.5	1	2	3	4	5	6	7	8	9	10	15
AAA	2	34	39	28	23	31	24	14	29	37	61	117
AA+	44	76	85	78	73	80	74	64	63	87	112	170
AA	84	116	124	116	111	118	110	99	96	119	142	192
AA-	124	156	166	161	161	173	162	147	141	160	180	213
A+	274	306	316	311	311	323	312	297	291	310	330	363
A	299	331	341	336	336	348	337	322	316	335	355	388
A-	374	406	416	411	411	423	412	397	391	410	430	463
BBB+	399	431	441	436	436	448	437	422	416	435	455	488
BBB	424	456	466	461	461	473	462	447	441	460	480	513
BBB-	474	506	516	511	511	523	512	497	491	510	530	563

Source:
<https://www.fimmda.org/modules/content/?p=1031>

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Estimating Cost of Debt from Coupon of a Recently Issued Bond

Issuer	Rating(s)	Coupon / Yield
Power Finance Corporation	CRISIL AAA / ICRA AAA	7.30%
Rural Electrification Corp (REC)	CRISIL AAA / ICRA AAA	7.25%
Housing & Urban Dev. Corp (HUDCO)	CARE AAA / ICRA AAA	7.20%
Indian Railway Finance Corp (IRFC)	CRISIL AAA / ICRA AAA	7.15%
Mahindra & Mahindra Financial Services	CRISIL AAA / ICRA AAA	7.90%
L&T Finance	CRISIL AAA / ICRA AAA	8.10%
Mufin Green Finance	CARE BBB+	11.75%
Refyne Finance	CARE BBB	12.00%
Unigold Finance	CARE BBB	11.50%
Shrivation Upscale Spaces (Real Estate)	CARE BB+	14.50%
Richfield Financial Services	CARE BB	15.00%

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Estimating Cost of Debt from Yield

Normal Market is Open As of 17-Jul-2026 11:09 IST Advances - 27 | Declines - 23 | Unchanged - 565 [Download \(.csv\)](#)

Value Convention ☐ Lakhs ☒ Crores ☐ Billions

SYMBOL	SERIES	ISIN	COUPON RATE	FACE VALUE	LTP	%CHNG	VOLUME (Shares)	VALUE (₹ Crores)	CREDIT RATING AGENCY	CREDIT RATING	MATURITY DATE
10JFL28	N4	INE477L08147	10	1,000.00	999.00	-7.84	1,117	0.11	CRISIL, BRCK	AA/Stable, AA+/Negative	03-Nov-2028
865IRFC29	NA	INE053F07686	8.65	1,000.00	1,097.00	-0.27	501	0.05	-	-	18-Feb-2029
SCLO1825	N6	INE148I07XX3	0.0	1,000.00	1,058.50	-0.05	380	0.04	CRISIL, ICRA	AA/Stable, AA/Stable	01-Aug-2027
975NHAI29	N6	INE906807DF8	8.75	1,000.00	1,107.00	0.21	310	0.03	-	-	05-Feb-2029
915SCL26	NE	INE148I08256	9.15	1,000.00	1,059.20	-1.01	320	0.03	CARE	AAA/	26-Sep-2026
865LIT26	NN	INE027E07BC8	8.65	1,000.00	1,039.12	-1.07	207	0.02	FITCH	AA+/Stable	23-Dec-2026
852HFL29	Y5	INE530B07591	8.52	1,000.00	990.00	0.00	145	0.01	CRISIL, BRICKWORKS	AA/Stable, AA+/Stable	06-Mar-2029
96HFL28A	NF	INE530B08102	9.6	1,000.00	1,001.00	0.00	125	0.01	CRISIL	AA/Negative	24-Jun-2028
897MOFSL29	N4	INE338I07164	8.97	1,000.00	1,024.00	0.89	105	0.01	CRISIL	AA/Stable	09-May-2029
903SFL28	Y1	INE721A07NUI	9.03	1,000.00	1,046.00	0.00	100	0.01	CRISIL, FICH	AA+/Stable, AA+/Stable	12-Jul-2028
865HFL28	N5	INE530B07310	8.65	1,000.00	990.00	-0.09	100	0.01	CRISIL, ICRA	AA/Stable, AA/Stable	24-Jan-2028
11FL2C28C	N6	INE477L08162	0	1,000.00	1,564.30	-0.24	44	0.01	CRISIL, BRCK	AA/Stable, AA+/Negative	03-Nov-2028
10CAGL27	N5	INE741K07496	10.0	1,000.00	1,020.00	0.39	60	0.01	FITCH	AA-/Stable	23-Nov-2027

Source: <https://www.nseindia.com/market-data/bonds-traded-in-capital-market>

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Estimating Cost of Debt of Titan

EQUITY AND LIABILITIES			
Equity			
(a) Equity share capital	13.1	89	89
(b) Other equity	13.2	20,358	16,722
	TOTAL EQUITY	20,487	16,811
Liabilities			
(1) Non-current liabilities			
(a) Financial liabilities			
(i) Borrowings	14.1	-	420
(ii) Lease liabilities	14.2	2,142	1,943
(b) Provisions	15	213	256
		2,355	2,619
(2) Current liabilities			
(a) Financial liabilities			
(i) Borrowings	16.1	6,239	7,483
(ii) Gold on loan	16.2	14,314	7,043
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- Total outstanding dues of micro and small enterprises	16.4	531	158
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(b) Other current liabilities	17	5,466	4,069
(c) Provisions	18	269	130
(d) Current tax liabilities (net)	8	14	32
		30,775	21,645
TOTAL EQUITY AND LIABILITIES		53,617	41,075

23

What is the total debt of Titan?

- You can divide the debt of a non-financial company into interest-bearing and non-interest-bearing liabilities.
- What are the interest-bearing liabilities?
- What are non-interest-bearing liabilities?
- Are the non-interest-bearing liabilities really free?

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What to do with non-interest-bearing liabilities?

- Assume that a company buys raw materials and gets 1 year credit on it. It needs to pay Rs.1100 crores.
- Assume that its cost of debt is 10%.
- Should we treat the trade payable as a source of fund while estimating the cost of debt (and capital)?

25

What if there are different debt instruments with different costs of debt?

- XYZ Limited has issued two types of debt at 12% (Rs.100 million) and 14% cost (Rs.200 million) respectively.
- Projected EBIT = Rs.100 million
- Tax Rate = 30%
- $K_e = 20\%$
- $g = 0$
- What is the value of XYZ?

26

Three Choices...

- We can directly value the equity using the FCFE method
- We can use only one (or more) of the two (or more) types of debt issued by the company while finding the cost of capital
- We can use all the debt instruments issued by the company while finding the cost of capital.

27

Consistency Issues

- If D_1 is a source of capital:
 - Do not subtract interest on D_1 while computing FCF
 - Subtract D_1 from Enterprise Value, while computing Equity.
- If D_0 is not a source of capital:
 - Do subtract interest on D_0 while computing FCF
 - Add increase in D_0 , while computing FCF
 - Do not subtract D_0 from Enterprise Value, while computing Equity.

28

Cost of Foreign Currency Loans...

- Assume that a company has borrowed US\$ 10,000 @6% per annum.
- Suppose the current rupee-dollar spot rate is Rs.64/1\$. The 1-year forward rate is Rs.66/1\$.
 - Cost of debt if loan is repaid in 1 year
 - Cost of debt if loan is repaid in 5 years

29

Rupee Cost of 5-year Foreign Currency Loan

Year	0	1	2	3	4	5
Loan Amount	\$10,000					
Interest Payments		\$600	\$600	\$600	\$600	\$600
Principal Repayment						\$10,000
Expected Spot Rate	64.00	67.80	71.83	76.10	80.62	85.41
Rupee Interest Rate	7%					
U.S. \$ Interest Rate	1%					
Overall \$ Cash Flow	\$10,000.00	(\$600.00)	(\$600.00)	(\$600.00)	(\$600.00)	(\$10,600.00)
Rupee Cash Flow	\$640,000.00	(\$40,681.19)	(\$43,097.89)	(\$45,658.16)	(\$48,370.53)	(\$905,311.13)
Implied Rupee Cost of Debt	12.30%					

30

What if cash is present?

- Assume the following for a zero-growth company:
- EBIT = 100
 - Does not include interest income
- Debt = 100 (@10%)
- Cash = 50 (@7%)
- Tax rate = 30%
- $K_e = 20\%$
- Equity Valuation using FCFE Method:
- PBT = 93.5
- Net Income = 65.45
- Equity = 327.25

31

What if cash is present?

Cash is non-operating

- FCF = 70
- EV = 377.25 (= 327.25 + 100 – 50)
- WACC = 18.55%
 - Effective cost of debt is 13%
- EV (check) = $70 / 18.55\% = 377.25$

Cash is operating

- FCF = 72.45
(= $(100 + 3.5) \times 0.7$)
- EV = 327.5 + 100 = 427.5
- WACC = 16.957%
- EV (check) = $72.45 / 16.957\% = 427.25$

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Cost of Debt at Titan

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Credit Rating of Titan

Instrument	ICRA	CRISIL	Brickwork
Non Fund Based Limits	A1+	A1+	-
Fund Based Limits	AAA (Stable)	AAA (Stable)	AAA (Stable)
Fixed Deposit Programme	MAAA (Stable)	-	-
Commercial paper	A1+	-	A1+
Gold On Lease Limits	-	-	A1+

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Recent Borrowings by Titan

Instrument	Amount	Rating(s)	Coupon / Yield	Date / Tenor
Commercial Paper (CP)	₹1,000 Cr	CARE A1+	Issued at ₹4,92,794.50 per ₹5,00,000 face value → effective yield ~5.8–6%	Listed Dec 2025, matured Mar 2026 rhnvm.github.io
Commercial Paper (CP)	₹5,000 Cr	CARE A1+	Similar short-term yields (~6%)	Rating reaffirmed May 2026 CARE Ratings
Bank Facilities (Working Capital, Term Loans)	~₹16,500 Cr	CARE AAA (Stable) / CARE A1+	Linked to prevailing bank lending rates (not disclosed publicly, but typically 8–9% for AAA corporates)	Reaffirmed May 2026 CARE Ratings
Fixed Deposit Programme	₹3,000 Cr	ICRA AAA (Stable)	Retail FD rates ~6.5–7% depending on tenor	Rating reaffirmed July 2026 nsearchives.nseind...

35

Cost of Equity

36

CAPM

- $E(r_i) = rf + \beta \times (E(r_m) - rf)$
 - Fama, French (1992) find beta not to explain stock returns.
 - Martingale Asset Management recommends investing in low-risk stocks.
- You need to know how to estimate
 - **The risk-free rate**
 - **Market risk premium**
 - **Beta**
- Alternatively, use Multi-factor Models.
 - Arbitrage Pricing Theory (You need to estimate the factor premiums)
 - Fama-French Multi-factor Models (Popular in industry, data easily available)

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Risk-free rate (17 July 2026)

Date	Tenor Bucket	Security	YTM (%)
16-07-2026	91D	91 DTB (16/10/2026)	5.3324
16-07-2026	182D	182 DTB (14/01/2027)	5.5693
16-07-2026	364D	364 DTB (16/07/2027)	5.7197
16-07-2026	1Y-2Y	7.06% GS 2028	5.95
16-07-2026	4Y-5Y	6.36% GS 2031	6.4331
16-07-2026	9Y-10Y	6.94% GS 2036	6.7498
16-07-2026	13Y-15Y	6.68% GS 2040	6.971
16-07-2026	28Y-30Y	7.24% GS 2055	7.3737
16-07-2026	5Y	7.07% WEST BENGAL SGS 2031	7.07
16-07-2026	10Y	7.51% ANDHRA PRADESH SGS 2036	7.51

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India's 5-Year CDS

India Government Bonds - Yields Curve

COUNTRY • SUMMARY	
	INDIA
Last Update: 17 Jul 2026, 2:15 GMT+0	
Main Indicators	
10-Year Gov.Bond Yield: 6.751%	S&P Rating: BBB
Spread vs 2-Year Bond: 80.5 bp	
Central Bank Rate: 5.25% (December 2025)	5Y Credit Default Swap: 87.67 bp (1.46% implied probability of default)

Source: <https://www.worldgovernmentbonds.com/country/india/>

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Risk-free Rate

- If the yield curve is not flat, how to find R_f ?

Year	1	2	3
Cash Flows	100	100	200

Year	1	2	3
Spot Rate	5%	6%	7%

$$Value = \frac{100}{1.05} + \frac{100}{1.06^2} + \frac{200}{1.07^3} = 347.50$$

$$347.50 = \frac{100}{1+r} + \frac{100}{(1+r)^2} + \frac{200}{(1+r)^3}$$

$$\Rightarrow r = 6.52\%$$

40

Yield for different cashflow patterns

Year	1	2	3
Cashflows	100	10	10
spot rate	5%	6%	7%
yield	5.49%		

Year	1	2	3
Cashflows	100	100	100
spot rate	5%	6%	7%
yield	6.29%		

Year	1	2	3
Cashflows	100	100	1000
spot rate	5%	6%	7%
yield	6.87%		

Year	1	2	3
Cashflows	100	100	10000
spot rate	5%	6%	7%
yield	6.99%		

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Market Risk Premium

TABLE 7.1A Average rates of return on short-term G-Sec Yields, AAA-rated corporate bonds and common stocks, 1978 – 2013 in India (figures in percent per year)

	Average Annual Rate of Return		Average Risk Premium (Extra Return versus 1-Year G-Sec Yields)
	Nominal	Real	
1-Year G-Sec Yield	9.22	1.55	0.00
AAA-Rated Corporate Bonds	13.42	5.75	4.20
Common Stocks	21.14	13.48	11.92

Source: Compiled from www.rbi.org.in

Time Period Covered in this Release

release	Start.Month	End.Month
2025-12	October 1993	December 2025

Source:
<https://faculty.iima.ac.in/iffm/Indian-Fama-French-Momentum/>

Factor Returns

	Last month	Last 3 months	Year to date	Since 01-Jan-1994
Market premium %	-8.17	4.53	-4.89	4.93
SMB %	-35.25	-31.70	-19.59	-3.71
HML %	24.15	6.46	11.38	6.61
WML %	16.46	37.09	-5.53	10.65

All returns are *annualized* geometric mean returns

Country	Total Equity Risk Premium
India	7.08%

Source:
https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/ctryprem.html

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Getting Data from IIMA

- Source: [Fama French and Momentum Factors: Data Library for Indian Market \(iima.ac.in\)](http://iima.ac.in)

Survivorship-Bias Adjusted (Recommended)

- Fama French and Momentum Factors:
[Daily](#) [Monthly](#) [Yearly](#)
- Six Portfolios formed on Size and Value:
[Daily](#) [Monthly](#) [Yearly](#)

43

Estimating Beta...

$$\beta = \frac{\text{Covariance}(R_i, R_m)}{\text{Variance}(R_m)} = \frac{\sigma_{i,m}}{\sigma_m^2} \dots (17)$$

- How to compute returns

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Computing Returns

- Arithmetic or logarithmic
- Use closing price or adjusted price?
- Adjustments for
 - Dividends
 - Bonus
 - Split

45

Rights issue

- In 2021, RIL made a rights issue in the ratio of 1:15 to its shareholders to subscribe to the rights shares at Rs.1257. The stock price on the record date was Rs.1435.

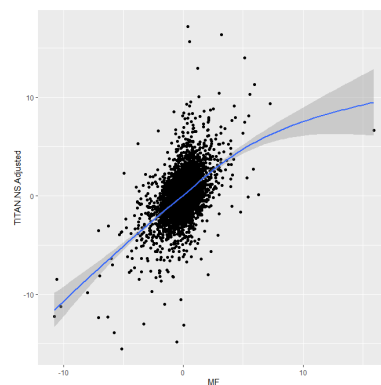
46

Estimating Beta

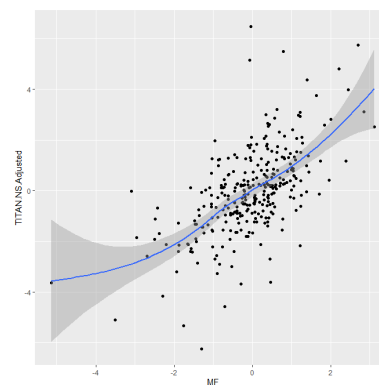
- What should be the estimation period...
 - 1-year, 2-years, 5-years?
- How frequently should we compute returns?
 - Daily, weekly, monthly?
- Keep the following two principles in mind:
 - The operating and financing risk of the company should not have changed during the estimation period.
 - A larger time period is better

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Effect of Estimation Period



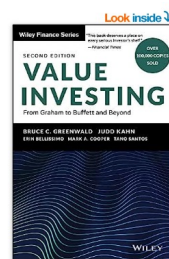
Beta = 0.998



Beta = 0.996

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Prof. Bruce Greenwald of Columbia



Value Investing: From Graham to Buffett Hardcover – Illustrated, 8 December 2020

by Bruce C. Greenwald (Author), Judd Kahn (Author), Erin Bellissimo (Author), Mark A. Cooper (Author), Tano Santos (Author)

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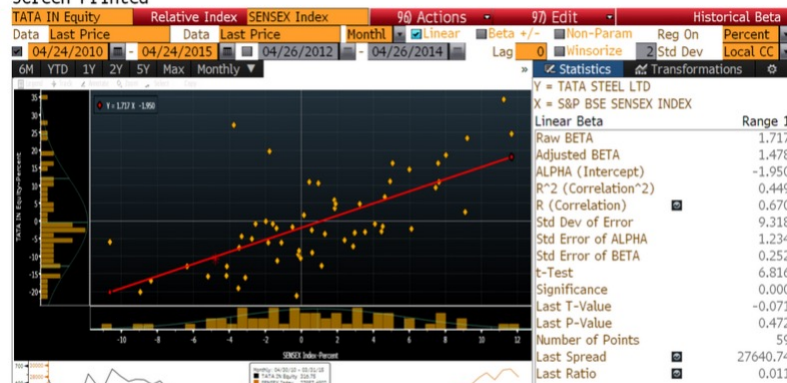
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No Cost EMI: Avail No Cost EMI on select cards for orders above ₹3000 [\[Details\]](#)

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Beta of Tata Steel

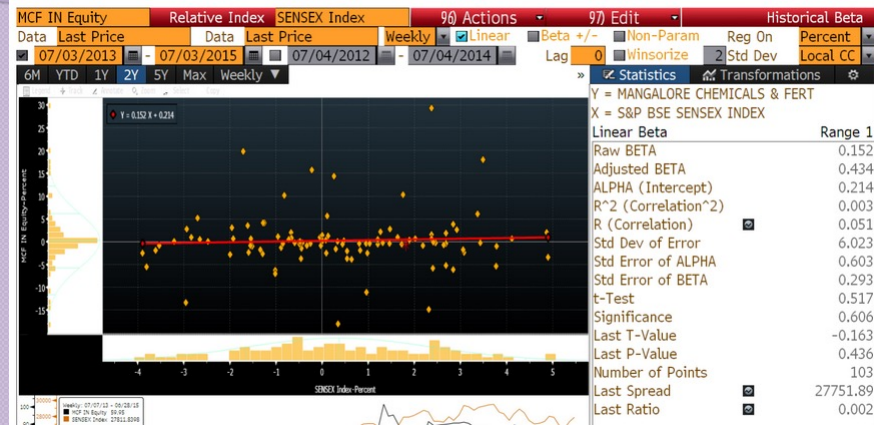
Screen Printed



Beta increases from 1.348 to 1.717 when monthly returns for the last 60 months are used.

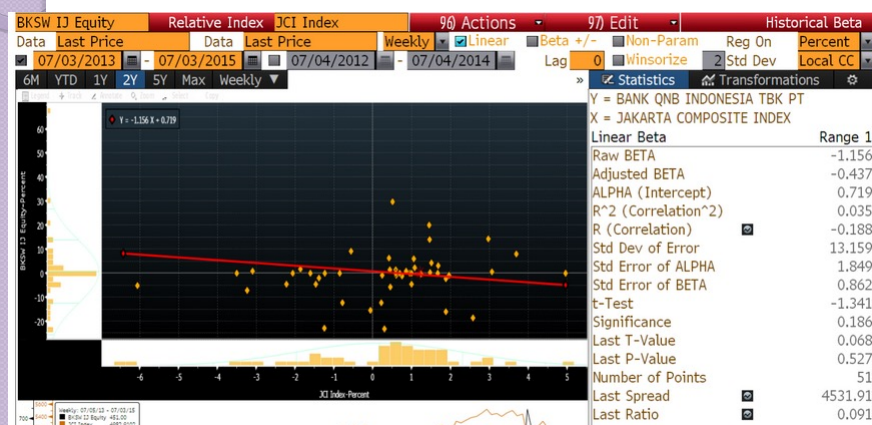
50

Low Beta (beta of MCF)



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Negative Beta (beta of Bank QNB Indonesia)



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Cost of Warrants

- What is a warrant?
- First estimate the beta of a call option.
 - The beta of a stock is 1.2. You invest 50% of your money in the stock and the remaining 50% in risk-free assets. What is the beta of your portfolio?
 - You borrow 50% money and invest 150% in the stock. What is the beta of your portfolio?
- Then estimate the beta of the warrant.
- Then estimate the cost of the warrant.

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